

Business Problem Statement

A leading banking institution aims to gain deeper insights into its customer base to improve profitability, customer retention, and personalized financial services. The management team wants to analyze customer demographics, banking relationships, income levels, deposits, loans, credit card usage, savings behavior, and risk profiles to identify key customer segments and financial trends.

The primary business question is:

"How can the bank leverage customer banking data to identify high-value customers, optimize product offerings, manage risk effectively, and enhance customer loyalty and engagement?"

The analysis will focus on uncovering patterns in customer deposits, loans, savings accounts, credit card balances, business lending, and investment preferences to support data-driven decision-making and strategic growth initiatives.

Deliverables

1. **Data Preparation & Modeling (Python):** Clean, transform, and preprocess the banking customer dataset. Handle missing values, create relevant features, and prepare the data for analysis.
2. **Data Analysis (SQL):** Structure the banking data into relational tables and perform SQL queries to analyze customer demographics, deposits, loans, credit card usage, savings behavior, and customer profitability.
3. **Visualization & Insights (Power BI):** Develop an interactive dashboard that highlights key banking KPIs, customer segments, loan and deposit trends, risk distribution, and relationship banking performance to support strategic decision-making.
4. **Report and Presentation:** Prepare a comprehensive report summarizing analytical findings, business insights, and recommendations. Create a presentation to communicate customer trends, financial performance, and growth opportunities to stakeholders.
5. **GitHub Repository:** Maintain a well-organized repository containing Python scripts, SQL queries, data documentation, dashboard files, reports, and project resources for reproducibility and collaboration.